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First Geoscience Outlook from the Pilot Well Drilled for Harnessing Deep Geothermal Energy in Vienna

Kanchan Dasgupta, Rafal Skupio, Thomas Gumpenberger, Barbara Stummer, Astrid Metz, Wolfgang Hujer, and Elias Mekonnen, OMV Exploration & Production GmbH

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Geothermal energy makes use of the heat beneath the earth's surface to provide a renewable source of heat and power. As a sustainable alternative to fossil fuels, it can help to accelerate the energy transition and reduce carbon emissions. Geothermal energy has a low carbon and areal footprint, it reduces dependence on energy imports and fossil fuels, and provides a constant, secure, and nearly limitless supply of heat, promoting regional economic growth. In case of hydrothermal systems, hot formation water is pumped to the surface, where its heat is captured using heat exchangers and fed into the district heating network. The cooled water is reinjected to the reservoir where it is reheated by the surrounding rocks. When water temperatures are higher than 140°C, the energy can also be used for power generation. The city of Vienna aims to make its entire district heating grid carbon-neutral by 2040. According to the ambitious plans, more than 200 MWth of thermal power should be produced in the next decade in or close to the City of Vienna (Keglovic and Novotny 2023). To help reach this goal, a joint venture (deeper Tiefengeothermie GmbH) has been established in 2023 between OMV and Wien Energie.

In the Vienna basin different geothermal systems exist – details can be found in [Sachsenhofer et al \(2024\)](#). The following is a summary of the geothermal potential in the Vienna Basin modified after in [Sachsenhofer et al \(2024\)](#). According to [Goldbrunner and Goetzl \(2019\)](#), Vienna Basin's principal formations that could support geothermal energy exploitation are the Middle and Upper Triassic carbonates in the Calcareous Alps (Hauptdolomit Formation), which occur in the depth range of 3000 to 5000 m, and Miocene 'Aderklaa conglomerates' (Rothneusiedl Formation) in the depth range of 2000 to 3500 m. In case of the latter formation, through the interdisciplinary 'GeoTief Wien' research project and OMV internal geothermal potential evaluation, different geological aspects (reservoir parameters, structural modelling, geomechanics, etc.) were studied, and based on a high-resolution 3D seismic survey, structural investigations were performed ([Schreilechner et al. 2022](#); [Dasgupta et al. 2023](#)). Additionally, a production test was completed in 2021 in the Rothneusiedl Formation, which produced water with a temperature of 86° C at the wellhead from a perforation interval from 2640 to 2700 m depth (MD), confirming suitable reservoir temperatures for geothermal usage ([Sachsenhofer et al. 2024](#)). It also provided valuable information on the chemistry of the saline aquifer. Although well tests in the Hauptdolomit Formation indicated higher flow rates and higher temperatures, it was decided to develop geothermal projects in the Rothneusiedl Formation first, as

this has lower geological and economic risks (Stern 2023; Siedl 2024). The first project in relation to this is a geothermal plant in Aspern, northeastern part of Vienna. The plant aims to have a capacity of 20 MW in combination with heat pumps, which will be enough to supply around 20,000 households, resulting in potential CO₂ savings of up to 54,000 metric tons per year. The final goal is to establish up to 7 plants with a combined capacity of 200 MW - enough to heat up to 200,000 homes.

As part of this project, a pilot well and a first geothermal doublet (injector/producer) are drilled. These deep wells reach depths of over 3,000 meters to use the hot formation water for heat generation. The pilot well provided core data and well-log data from the Rothneusiedl Formation and serves as monitoring station for seismicity and thermal break-through indicator as it is located between the injector and producer (Knezevic et al. 2023). This study presents the first-hand, preliminary results obtained from the pilot well and compares the understanding gathered pre-drill with the post-drilling results.

Introduction

Geothermal energy makes use of the heat beneath the earth's surface to provide a renewable source of heat and power. As a sustainable alternative to fossil fuels, it can help to accelerate the energy transition and reduce carbon emissions. Geothermal energy has a low carbon and areal footprint, it reduces dependence on energy imports and fossil fuels, and provides a constant, secure, and nearly limitless supply of heat, promoting regional economic growth. In case of hydrothermal systems, hot formation water is pumped to the surface, where its heat is captured using heat exchangers and fed into the district heating network. The cooled water is reinjected to the reservoir where it is reheated by the surrounding rocks. When water temperatures are higher than 140°C, the energy can also be used for power generation. The city of Vienna aims to make its entire district heating grid carbon-neutral by 2040. According to the ambitious plans, more than 200 MWth of thermal power should be produced in the next decade in or close to the City of Vienna (Keglovic and Novotny 2023). To help reach this goal, a joint venture (deeep Tiefengeothermie GmbH) has been established in 2023 between OMV and Wien Energie.

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Workflow

In this study geoscience workflows involving cores and logs have been innovatively modified to suit geothermal energy exploitation. All the cores were subjected to several analytical techniques sequentially, prior or simultaneous to wireline logs interpretation (Fig. 1). Once the whole cores were cleaned with clothes and tap water, a quick-look lithological description while within the core sleeves were done. This formed the basis of targeted sampling for the following analysis: spectral gamma-ray, CT-scan, X-ray diffraction, whole-core porosity, triaxial testing, routine core analysis and petrography. Following the preliminary sample selection, detailed sedimentological descriptions and CoreDNA testing using an in-house Wombat core analyser were carried out, simultaneously with wireline log interpretations.

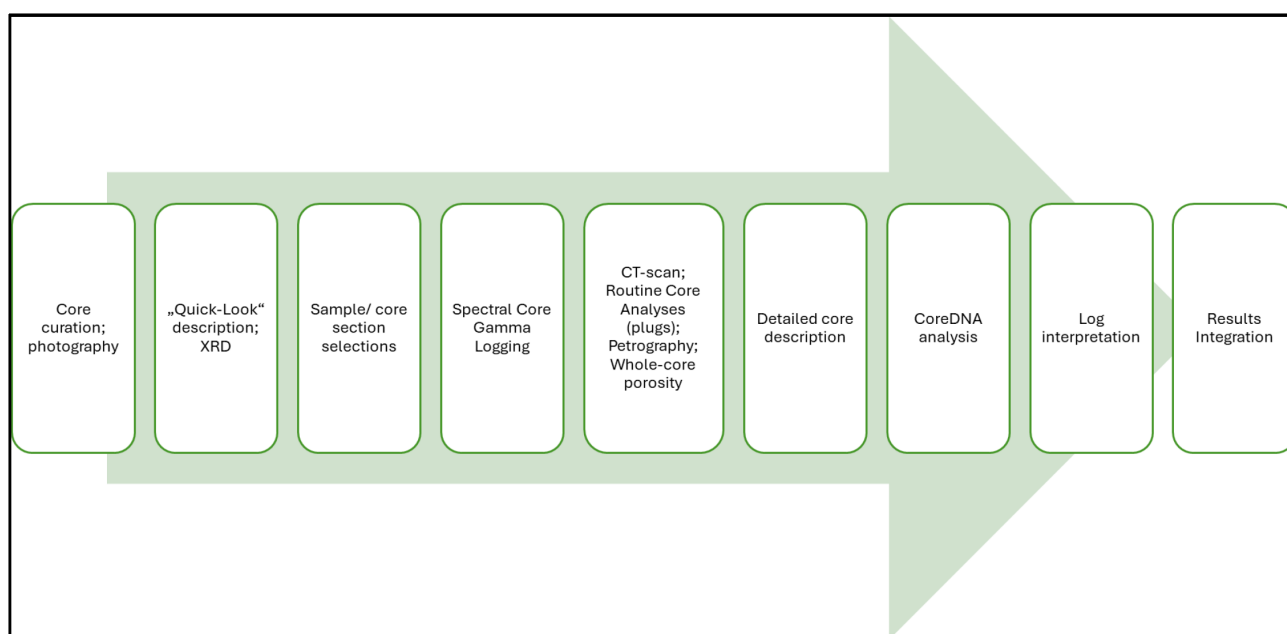


Figure 1—Study workflow and the analyses undertaken on the available cores.

Data acquisition and challenges

The unslabbed cores were laid out after removing them from their liners for macroscopic description in OMV Tech Center Lab in Gänserndorf, Austria. The typical core handling and processing steps were adjusted to accommodate the large diameter of the cores (13cm) and additional weights. The freshly cut cores were cleaned multiple times to remove mud filtrates and evaporated salts. Note that the cores were drilled with saline drilling mud. Digital graphic description was undertaken using EasyCore software. All unslabbed cores were also ‘photographed’ with a setup that turns a customary smartphone into a semi-automatic core scanner utilising its panoramic photo function while guided on a rail (Gegg and Gegg 2023).

Typical *sedimentological core description* technique utilises the Udden–Wentworth scale of grain size (clay, silt, sand, etc) for objective description of clastic sediment but it inadequately covers gravel, the dominant sediment size fraction in this study. However, Blair and McPherson (1999) extended the Udden–Wentworth scale to cover the coarser realm of sedimentary particles, including various grades of pebbles,

cobbles, and boulders in the gravel fraction, and various grades of blocks, slabs, monoliths, and megaliths in the megagravel fraction.

Spectral gamma-ray measurements (SCGL) were conducted using a handheld GT-40 spectrometer equipped with a 2×2" BGO detector. Although the device is primarily designed for surface radiation measurements, it was adapted for core logging through additional calibration and the use of a lead collimator. Logging was carried out at 20 cm intervals, with an acquisition time of 2 minutes per measurement. The acquired spectra were analyzed to produce calibrated logs of potassium (K [%]), uranium (U [ppm]), thorium (Th [ppm]), and total gamma-ray (GR) in API units across the entire cored interval. The total GR data enabled precise depth matching between core and well logs and provided improved resolution of radiation level variations along the profile.

The *bulk mineralogical composition* was determined using a Bruker AXS D8 Advance X-ray diffraction spectrometer (copper radiation generated at 40 kV and 40 mA). The software program DIFFRAC.EVA V3 was used for identification of different mineral phases. Quantification of minerals detected by XRD is based on peak heights within the spectrum based on the method of Schultz (1964) and internal standards. Based on comparison with other methods and other laboratories, the quantification results show an absolute failure range of 2-5%.

Standard *petrographical techniques* were used throughout the thin-section analysis program. All thin-section samples were vacuum impregnated with blue-dyed epoxy resin to enable the characterization of the pore spaces. Thin-sections were also stained for carbonates with combined alizarin red-S and potassium ferricyanide to facilitate the identification of the carbonate mineralogy (Dickson, 1965).

Whole-core porosity: whole-core sections of 20 cm length were selected to accommodate the gravel-size clasts of the samples. The samples partially exhibited salt precipitations from reservoir fluids and/or drilling mud. As the weight of the whole-core sections (typically 6 to 7 kg) exceed the capacity of the available Soxhlet apparatus, samples were cleaned via an ad-hoc devised method of soaking in methanol for one week. The fluid was exchanged two to three times during that period. While the silver nitrate tests performed at the end of the soaking period still indicated the presence of chlorides, the cleaning was nevertheless stopped after one week for time constraints. As well the volume of potentially remaining salt was calculated to be low enough not to influence the porosity results. The process did remove plenty of salt from the samples. When recovering the methanol salt residue was clearly visible in the round bottomed flasks. This process consumes copious amounts of methanol. For sustainability, the used methanol was distilled to remove dissolved salts and reused. Soaking in methanol also made it easier to remove any adhering drilling mud, which along with any loosely attached grains were removed by means of compressed air. *Grain volume* (V_{grain}) was recorded on a porosimeter based on Boyle's Law using helium as test gas. The sample chamber can accommodate core sections of up to 20 cm in length and 18 cm diameter. The calibration of the device was checked by means of aluminium cylinders of known volume. The relative error in V_{grain} was determined to be 1.6 %. For the resulting grain density (RHOG) the relative error was calculated as 2 %.

The biggest experimental challenge was the proper determination of the *bulk volume*. The samples were both too large and exhibited too large pores that Hg-immersion or saturation methods would consistently work. Bulk volume was thus determined by recording length and diameter to the nearest mm and using the formula for a right cylinder. For two samples the following corrections were made: a hole from plugging was assumed to be a right cylinder and a broken off part as a three-sided pyramid. These corrections are correct to some single digit cm^3 . Errors in sample diameter have a larger effect on bulk volume than errors in length as they enter the formula via the r^2 dependence. It was later noted that core sections with drilling induced fractures had marginally larger diameter (13.3 vs 13.0 cm) than other sections. This immediately resulted in a porosity that was several porosity units higher than adjacent sections. Otherwise, porosity values are correct to less than one porosity unit.

Routine core analysis: several 3.8 cm diameter plugs were drilled for routine core analysis. The lengths of majority of these plugs were 11 cm. Porosity and grain density are determined by means of the Helium (He) expansion method at ambient conditions (i.e., ambient temperature and without confining pressure). The device was designed and constructed in-house and allows for the simultaneous investigation of up to ten samples. The principle of operation is Boyle-Mariotte's law. Briefly, the sample is placed in the sample chamber and charged with approximately 2 bar(g) of He and given time to equilibrate at pressure p_1 . After that, the He is allowed to expand into a (He-filled) expansion chamber (originally at atmospheric pressure) and left to equilibrate at a lower pressure level p_2 . Prior knowledge of the sample and expansion chamber's volumes by measurement with steel plugs for calibration leads to the grain volume by solving following a set of gas equations for VGrain. Grain density (RHOG) follows straight from dividing sample mass m by VGrain. Typically, one steel plug is included in the measurement at alternating measurement positions as a quality control. In the current study, bulk volume was determined by calipers instead of Hg-buoyancy due to the presence of large pores in the samples.

The measurement of permeability requires the sample to be the shape of a right cylinder, i.e., a plug needs to be available, ideally with a ratio of length to diameter larger than two. For the measurements, the plugs are placed inside a Hassler-sleeve at a confining pressure of ~ 35 bar. The test-medium is gaseous nitrogen at either constant flowrate or constant inlet pressure, depending on sample permeability. Solving Darcy's law for compressible media yields permeability k . If sample shape was irregular (e.g. chipped during drilling) no permeability data was obtained (only RHOG).

All suitable core sections were also subjected to *CoreDNA analysis* (EPSLOG©) through an apparatus called Wombat, which combines a suite of non-destructive analyses, or scanning with multiple sensors, to produce a transdisciplinary, high-resolution dataset. This typically includes ultra-high-resolution pictures, portable XRF measurements, laser scan, probe permeability, P- and S-wave velocity log and rock strength data. The cores are scanned along a fit-for-purpose surface with a collection of sensors. The flat, horizontal surface is created by successive dry cuts with a square polycrystalline diamond compound blade. Each dry cut removes a sub-mm thick rock layer. All CoreDNA sensors are deployed on the same groove created by the blade. The dataset generated for this study includes high and ultra-high-resolution pictures, laser scan, portable XRF measurements, P- and S-wave velocity log and scratch test data.

For the ultra-high-resolution (UHR) pictures, photographs of the cores were taken continuously along the longitudinal axis with telecentric lenses, giving a resolution of less than $2\ \mu\text{m}$ per pixel. The portable XRF system reports elemental composition of the cores from 1-cm spot, however, it cannot differentiate elements lighter than Mg. A scratch test involves tracing a groove on the surface of a specimen with a cutting tool. The test has been widely used to study the cutting process and to assess the effect of rock properties and tool characteristics on the performance of drilling or excavation tools. Unconfined compressive strength (UCS) of rocks can be reliably assessed from scratch tests performed under controlled conditions. These include utilising a sharp cutter and depth of cut small enough to assure that cutting takes place in the ductile regime. This technique yields a continuous log of the rock strength and the spatial variation of the strength, which in turns provides not only invaluable insights on the inhomogeneity of the rock but also a context to the conventional meaning of the UCS.

Results

In the studied pilot well, 4 intervals are cored: 3 targeting the Miocene Rothneusiedl Fm and one from the overlying 'caprock' section. In total around 96 metres of cores are recovered, however, this study focuses primarily on the 92 metres of large diameters (13 cm) cores from the Rothneusiedl Fm. The intervals for coring were selected to capture the depositional cycles identified within the Rothneusiedl Fm in previous in-house studies (Dasgupta et al. 2023). Overall, the recovered cored captured around 40% of the formation in that location, where it is around 230m thick. The maximum thickness of Rothneusiedl Fm

previously documented from the Schwechat and Marchfeld depressions is about 370m (Harzhauser et al 2020; Dasgupta et al., 2023), however previously studied cored intervals represent a maximum of 10% of the entire strata in a single well location.

The cores from the Rothneusiedl Fm dominantly comprise clast-supported, poorly sorted, conglomerates with sub-rounded to sub-angular, gravel-sized (granules, pebbles, cobbles) clasts and structureless fabric. The clasts, derived from diverse parent rocks, are commonly not imbricated (chaotically arranged). The following types of clasts have been identified: dolomite, limestone, sandstone, siltstone, claystone, quartzite, schists and undifferentiated crystalline rocks. Overall, carbonate clasts dominate, with the dolomite and limestone clasts preserve the original depositional texture (microfacies) as evident under microscope.

Sandstones and mudrocks are also present among the cores but in minor abundance. The sedimentological logging workflow utilised a bed/sub-bed scale lithofacies scheme, based on primary texture, lithology, sedimentary structures, and, where relevant, any additional features (e.g. fractures). However, note that it did not consider sorting and cement content. A total of 13 lithofacies have been identified. The lithofacies do not have direct environmental implications but provide the building blocks for sedimentary process interpretation and represent the fundamental qualifiers for samples/plugs selected for further analyses (e.g. RCA).

The lithofacies have been further grouped into bedset to bedset stack-scale for obtaining genetically related facies which are likely related to the environment of deposition. The interpretation of facies or depositional elements is based on the nature, stacking patterns, depositional context and relative occurrence of lithofacies. Overall, different types of braided channel-fills, braided bars and overbank depositional environments are envisaged in the current study.

In comparison to previous studies, the pilot well cores capture more diverse lithofacies and depositional environments. For examples, previous studies did not report the occurrence of conglomerate with rip-up mudclasts and mudrock formed through debris flow processes. This could be due to a multiple of reasons: core coverage of the formation and location of the study well.

XRD results highlighted the dominance of dolomite mineral among the conglomerate. Other carbonate minerals include calcite and ankerite, both of which also occur as an authigenic (cement) phase under microscope.

The large dimension of the cores posed several challenges in terms of core handling and processing, particularly as the cores are typically of conglomerates and drilled with saline drilling mud. Acquiring reservoir property data for coarse pebble conglomerates proved difficult as standard plugs would not be representative. Nevertheless, several routine plugs and full core pieces were retrieved for porosity and permeability measurements as well as petrography. Full core porosity data are obtained on 16 full core sections. The porosities recorded from the plugs have a wider range, with higher porosity values locally caused by drilling induced fractures. Lower porosities are typically observed at the bottom of the cored interval. Horizontal permeabilities (to N₂) are measured only on the plugs, excluding the plugs that developed cracks/fractures during the sample preparation process. Additional analyses to obtain permeabilities on full core samples are ongoing, including brine permeabilities.

The *CoreDNA analysis* could not be conducted on the entire cores from the current study as the cores are prone to mechanical damage. Furthermore, due to the gravel-sized conglomerates, laser-scan based topography maps are not analysed for the purpose of obtaining textural information as grains too big for the UHR images. Results from XRF, probe permeability and rock strength are integrated where present. An overview of the results obtained from the diverse analytical techniques is illustrated in Figure 2.

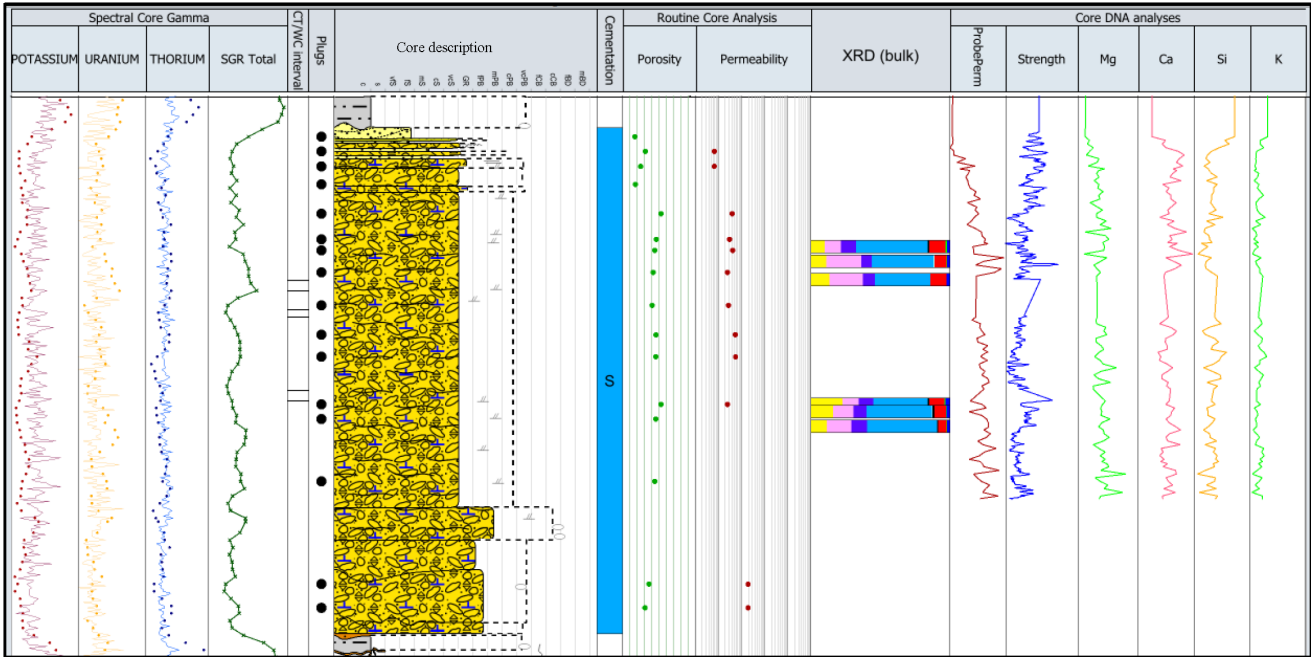


Figure 2—A schematic diagram illustrating some of the divers analyses undertaken on the cores in this study.

The petrophysical logging suite comprises gamma ray- caliper – neutron - density- photoelectric factor - sonic-resistivity. SGR measured on the cores are utilised for depth matching between core and log depths. The first and foremost task was to undertake lithology and porosity estimation, leading to poro-perm transformation and comparison with previous internal studies.

The logs within the Rothneusiedl Formation are of good to moderate quality. Log readings are very sensitive to borehole rugosity and breakouts due to the small diameter of the logging tools (compact well shuttle service; on drill string). All zones of faulty log readings are flagged as badhole for interpretation. Figure 3 plots an overview of the logging suite and interpretation. Log data confirms the predominance of dolomite with the data cloud plotting nearest to the dolomite line (Fig. 4). Initial interpretation of total porosity is based on density log as this is deemed the most reliable log in good hole conditions. The grain density is based on the mineralogical mixture obtained from XRD analysis, however, it appears a variable grain density (RHOG) might resolve the intervals with difference in reservoir quality better. Hence, a proper discriminator using more core data is necessary. Given the brine permeability assessment is currently undergoing, the poro-perm transform established during the previous study phases was agreed to serve as best estimate for a mid-case scenario.

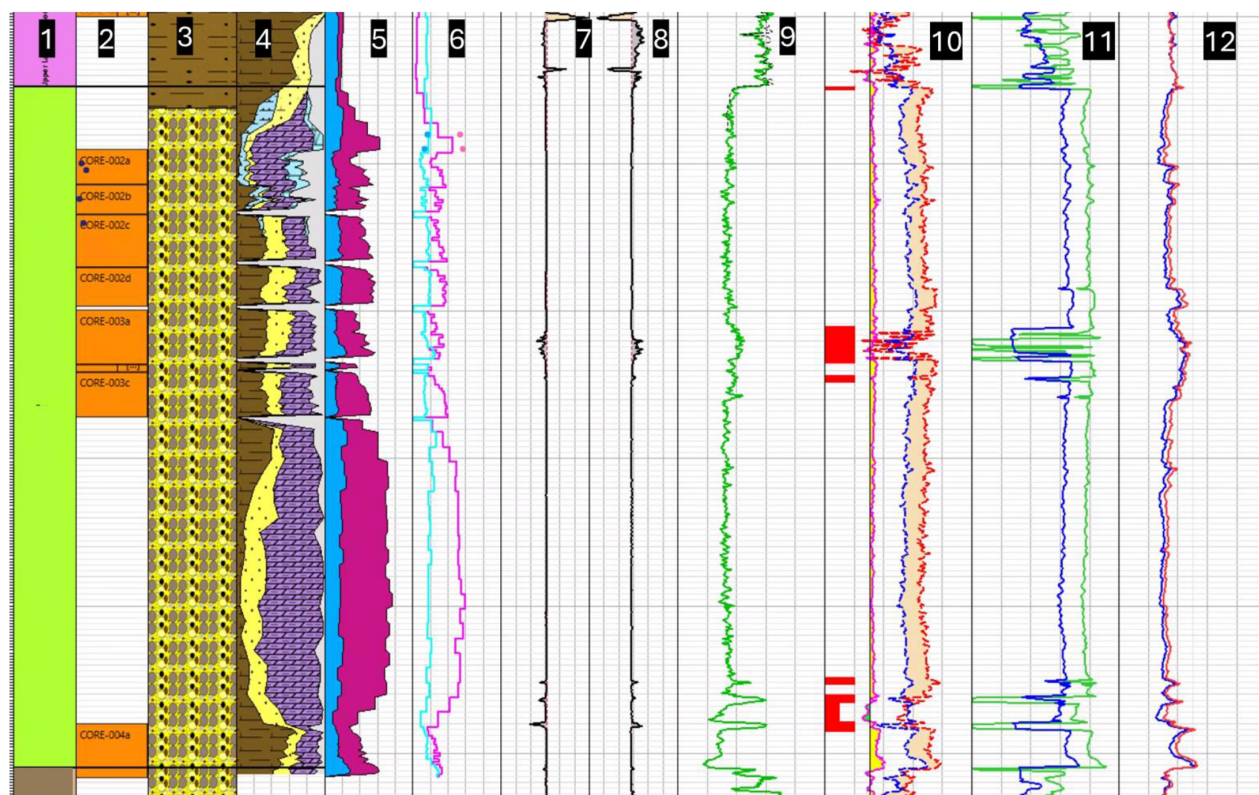


Figure 3—Overview of wireline logs collected in the pilot well. Track 1: Formation tops. Track 2: cored intervals. Tracks 3+4: Lithology from mudlog. Tracks 5+6: Calcimetry. Tracks 7+8: Caliper log. Track 9: GR, Track 10: Neutron – Density – PEF (red flag = badhole). Track 11: Sonic. Track 12: RES.

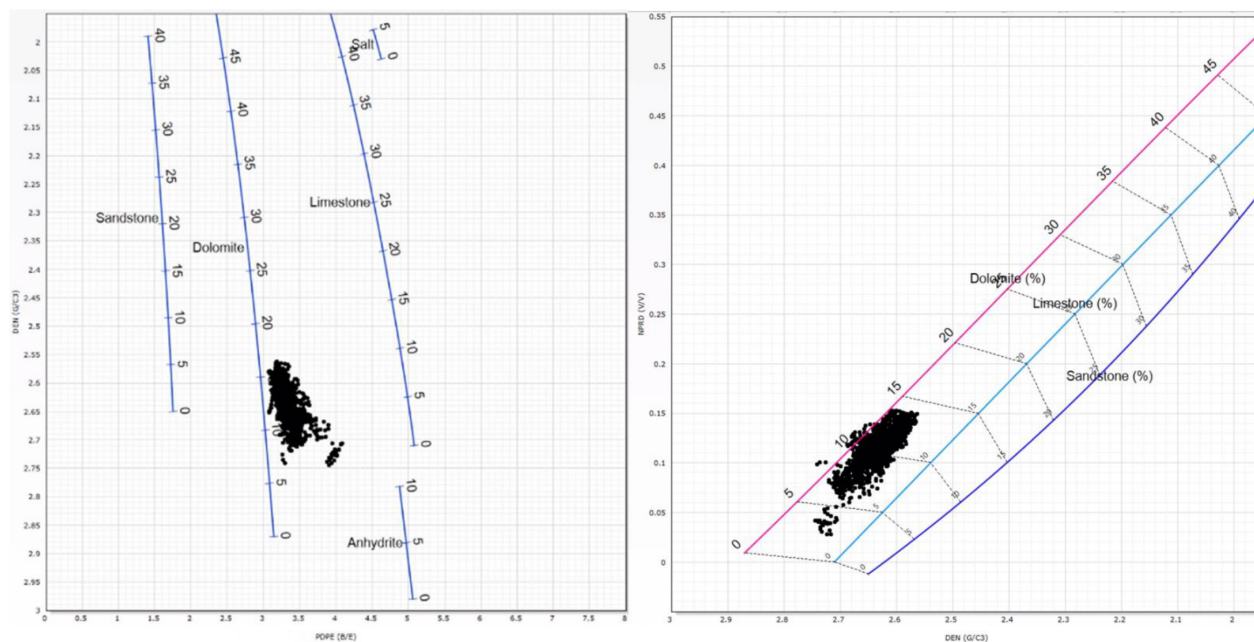


Figure 4—Lithology cross plots utilising log data (Left: DEN-PEF, right: NEUT- DEN).

Conclusion

The pilot well has provided core and log data from the Rothneusiedl Formation, informally known as Aderklaa Conglomerate in the Central Vienna Basin. Previous in-house studies calculated that this geological formation has the capacity to provide 200 000 Viennese households with renewable district

heating in 2030. This geoscience study presents a first-hand experience and results of exploiting deep geothermal energy within a city. It also compares results gathered during exploration phase with the pilot well data drilled for production. The lessons learned are also relevant for exploring conglomerate reservoirs for classical and low-carbon energy solutions elsewhere.

References

- Blair, T.C. and McPherson, J. 1999. Grain-size and textural classification of coarse sedimentary particles. *Journal of Sedimentary Research*, **69**, 6–19. [10.2110/jsr.69.6](https://doi.org/10.2110/jsr.69.6).
- Dasgupta, K., Gumpenberger, T., Stern, G., Siedl, W. and Biernat, M. 2023. Making the most out of historical cores for geothermal exploration. *Geological Society, London, Special Publications*, **527**, 399–412, <https://doi.org/10.1144/SP527-2022-9> Dasgupta et al 2023.
- Dickson J.A.D. 1965. A Modified Staining Technique for Carbonates in Thin Section. *Nature* **205** (4971), 587. <https://doi.org/10.1038/205587a0>.
- Gegg, L. and Gegg, J. 2023. Poor Man's Line Scan - a simple tool for the acquisition of high-resolution, undistorted drill core photos. *Scientific Drilling*, **32**, 55–59. [10.5194/sd-32-55-2023](https://doi.org/10.5194/sd-32-55-2023).
- Goldbrunner, J. and Goetzl, G. 2019. Geothermal Energy Use, Country Update for Austria. *European Geothermal Congress 2019*, 11–14 June 2019, Den Haag, The Netherlands. 1–10.
- Harzhauser, M., Kranner, M., Mandic, O., Strauss, P., Siedl, W. and Piller, W.E. 2020. Miocene lithostratigraphy of the northern and central Vienna Basin (Austria). *Austrian Journal of Earth Sciences*, **113**, 169–199, <https://doi.org/10.17738/ajes.2020.0011>
- Keglovic, P. and Novotny, B. 2023. Tiefengeothermie für die Dekarbonisierung der Fernwärme in Wien. Presented at ÖGEW/DGMK Herbstveranstaltung, November 2023, Vienna, Austria.
- Knezevic, R.N., Siedl, W., Stern, G. and Ebner, M. 2023. Application of Hydrocarbon Experience in fuel switch to geothermal energy: a Case study from Vienna Basin. 84th EAGE Annual Conference and Exhibition, 2023. 1–5, <https://doi.org/10.3997/2214-4609.202310589>
- Sachsenhofer, R., Misch, D., Rainer, T., Rupprecht, B. and Siedl, W. 2024. The Vienna Basin: Petroleum Systems, Storage and Geothermal Potential. *Geological Society, London, Special Publications*. 555. [10.1144/SP555-2023-205](https://doi.org/10.1144/SP555-2023-205).
- Schreilechner, M.G., Eichkitz, C.G. et al 2022. Interpretation and modelling for deep geothermal energy in Vienna. *First Break*, **40**, 90–99, <https://doi.org/10.3997/1365-2397.fb2022062>
- Schultz O. 2005. Catalogus fossilium Austriae. *Verl. der Österr. Akad. der Wiss, Wien*, 6911212 pp. <https://doi.org/10.1553/0x000d1cf8>.
- Siedl, W. 2024. Von einer regionalen geothermischen Machbarkeitsstudie hin zu einem ausgearbeiteten Bohrprojekt - Fallbeispiel Hydros Seestadt in Wien. Presented at GDMB Fachausschuss für Geothermie, March 2024, Vienna, Austria.
- Stern, G. 2023. Static subsurface risk assessment of geothermal projects. 84th EAGE Annual Conference & Exhibition, June 2023, Vienna, Austria, <https://doi.org/10.3997/2214-4609.2023101232>